Java Programming: Introduction and Fundamentals

Study notes for a document-based Retrieval-Augmented Generation (RAG) project

How to use these notes

These notes provide connected explanations, examples, terminology, and review questions. They are intended as a starter corpus for testing PDF extraction, chunking, embedding, and retrieval. Read the sections and ask questions whose answers appear in the material.

1. What is Java?

Java is a high-level, class-based programming language designed to be readable, portable, and suitable for a wide range of applications. Java source code is compiled into bytecode. The Java Virtual Machine (JVM) executes that bytecode, allowing the same compiled program to run on different operating systems when a compatible JVM is available.

2. JDK, JRE, and JVM

The Java Development Kit (JDK) contains tools for developing Java programs, including the compiler (javac), launcher, debugger, and runtime components. The Java Runtime Environment (JRE) is the runtime environment needed to run Java applications. The JVM is the virtual machine that loads, verifies, and executes bytecode. In modern Java distributions, the JRE is commonly delivered as part of a JDK rather than as a separate download.

3. A first Java program

A basic Java application can define a class with a main method. The main method is the conventional entry point used by the Java launcher.

Code: Hello World

```
public class HelloWorld {
public static void main(String[] args) {
System.out.println("Hello, Java!");
}
}
```

4. Variables and data types

A variable is a named storage location. Java is statically typed: a variable has a declared type, and the compiler checks many type errors before execution. Primitive types include byte, short, int, long, float, double, char, and boolean. Reference types include classes, arrays, interfaces, and enumerations.

5. Operators and expressions

Java provides arithmetic operators (+, -, *, /, %), comparison operators (==, !=, <, >, <=, >=), logical operators (&&, ||, !), assignment operators, and increment/decrement operators. Operator precedence determines grouping, but parentheses should be used when they make an expression easier to understand.

6. Decision-making

The if, else-if, and else statements allow a program to choose between paths. The switch statement is useful when one expression is compared with several alternatives. A switch expression can also produce a value in modern Java.

7. Loops

The for loop is commonly used when the number of iterations is known. The while loop repeats while a condition is true. The do-while loop checks its condition after executing the body, so its body runs at least once. Use break to exit a loop and continue to skip to the next iteration.

8. Arrays and strings

An array stores a fixed number of values of one type and uses zero-based indexing. Its length is available through the length field. String represents immutable text: operations that appear to modify a String create another String. StringBuilder is useful when many changes to text are needed.

9. Methods

A method groups statements into a reusable operation. A method declaration includes a return type, name, parameter list, and body. The void keyword indicates that a method does not return a value. Method overloading allows methods to share a name when their parameter lists differ.

10. Compilation and execution

Save source code in a .java file. The javac compiler translates it into bytecode in a .class file. The java launcher starts the JVM and runs the class. For example, compile with javac HelloWorld.java and run with java HelloWorld.

11. Good programming practices

Choose descriptive names, keep methods focused, use consistent indentation, validate input, and write comments that explain intent rather than restating obvious code. Compile frequently and test small pieces of functionality before combining them.

12. Review questions

1. What is bytecode?
2. What is the difference between JDK and JVM?
3. Name Java's eight primitive types.
4. How does a while loop differ from a do-while loop?
5. Why is String described as immutable?

End of document — Java study corpus